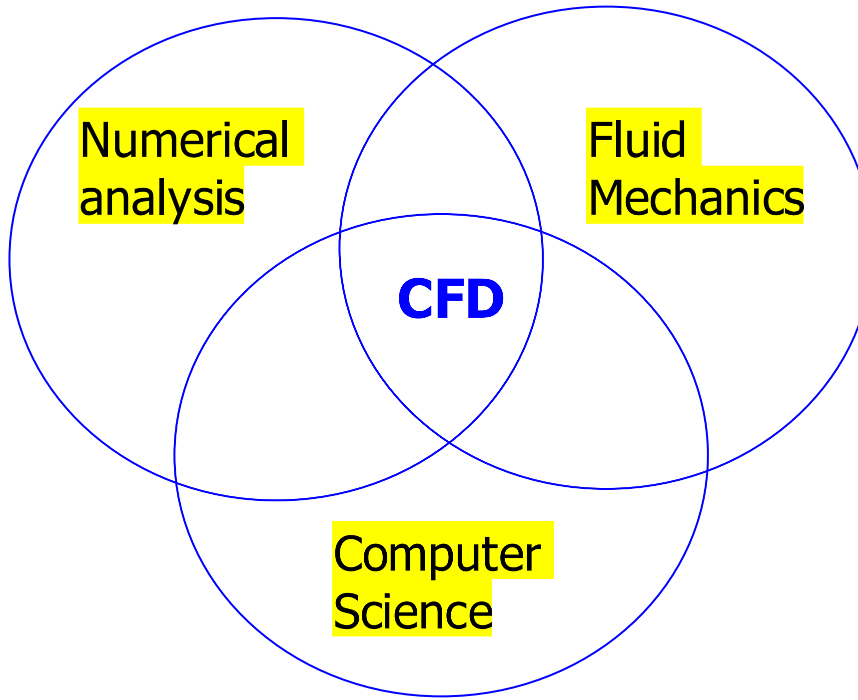
Introduction to partial differential equations in fluid mechanics: their types and some analytical solutions



Typical types of partial differential equations

Classification of Partial Differential Equations: consider the case of a second-order differential equation given in the general form as

$$A\frac{\partial^2\phi}{\partial x^2} + B\frac{\partial^2\phi}{\partial x\partial y} + C\frac{\partial^2\phi}{\partial y^2} + D\frac{\partial\phi}{\partial x} + E\frac{\partial\phi}{\partial y} + F\phi = G(x, y)$$

If the coefficients $A, B, C, D, E,$ and E are either constants or function of only (x, y) (do note contain ϕ or its derivative), the PDE is said to be a linear equation;

Otherwise, it is a nonlinear PDE.
The Navier – Stokes equation is a nonlinear PDE in general.

Classification of Partial Differential Equations:

$$A\frac{\partial^2\phi}{\partial x^2}+B\frac{\partial^2\phi}{\partial x\partial y}+C\frac{\partial^2\phi}{\partial y^2}+D\frac{\partial\phi}{\partial x}+E\frac{\partial\phi}{\partial y}+F\phi=G(x,y)$$

Parabolic PDE: $B^2-4AC=0$

Elliptic PDE: $B^2-4AC<0$

Hyperbolic PDE: $B^2-4AC>0$

Unsteady Navier-Stokes equations are elliptic in space and parabolic in time.

At steady state, the Navier-Stokes equations are elliptic.

In elliptic problems, the boundary conditions must be applied on all confining surfaces. These are boundary value problems.

Some examples

Heat transfer in a solid medium (no flow)

$$\rho\frac{\partial e}{\partial t}+\rho u_j\frac{\partial e}{\partial x_j}=-p\nabla\cdot\vec{u}+\rho\varepsilon+\frac{\partial}{\partial x_j}\left(k\frac{\partial T}{\partial x_j}\right)$$

When $\vec{u}=0$ $\longrightarrow$

$$\rho\frac{\partial e}{\partial t}=\frac{\partial}{\partial x_j}\left(k\frac{\partial T}{\partial x_j}\right)$$

$e=CT,\quad C$ is the specific heat

Assume: ρ, C, k are all constants, define thermal diffusivity $\alpha\equiv\frac{k}{\rho C}$

$$\frac{\partial T(\vec{x},t)}{\partial t}=\alpha\frac{\partial^2 T}{\partial x_j\partial x_j}$$

Special cases

1D transient heat transfer problem

$$\frac{\partial T(\vec{x},t)}{\partial t} = \alpha \frac{\partial^2 T}{\partial x_j \partial x_j}$$

$\rightarrow$

$$\frac{\partial T(x,t)}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}$$

Transient problem:
Solution depends on both the initial condition and boundary conditions

Parabolic PDE

2D steady-state heat transfer problem

$$\frac{\partial T(\vec{x},t)}{\partial t} = \alpha \frac{\partial^2 T}{\partial x_j \partial x_j}$$

$\rightarrow$

$$0 = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}$$ 2D Laplace equation

Equilibrium problem:
Solution is fully determined by the boundary conditions

Elliptic PDE

An example of analytical solution for elliptic PDE

$$0 = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}$$

2D Laplace equation

Physical domain: $0 \le x \le L, \quad 0 \le y \le L$

Boundary conditions: $T(x,0) = T_0$

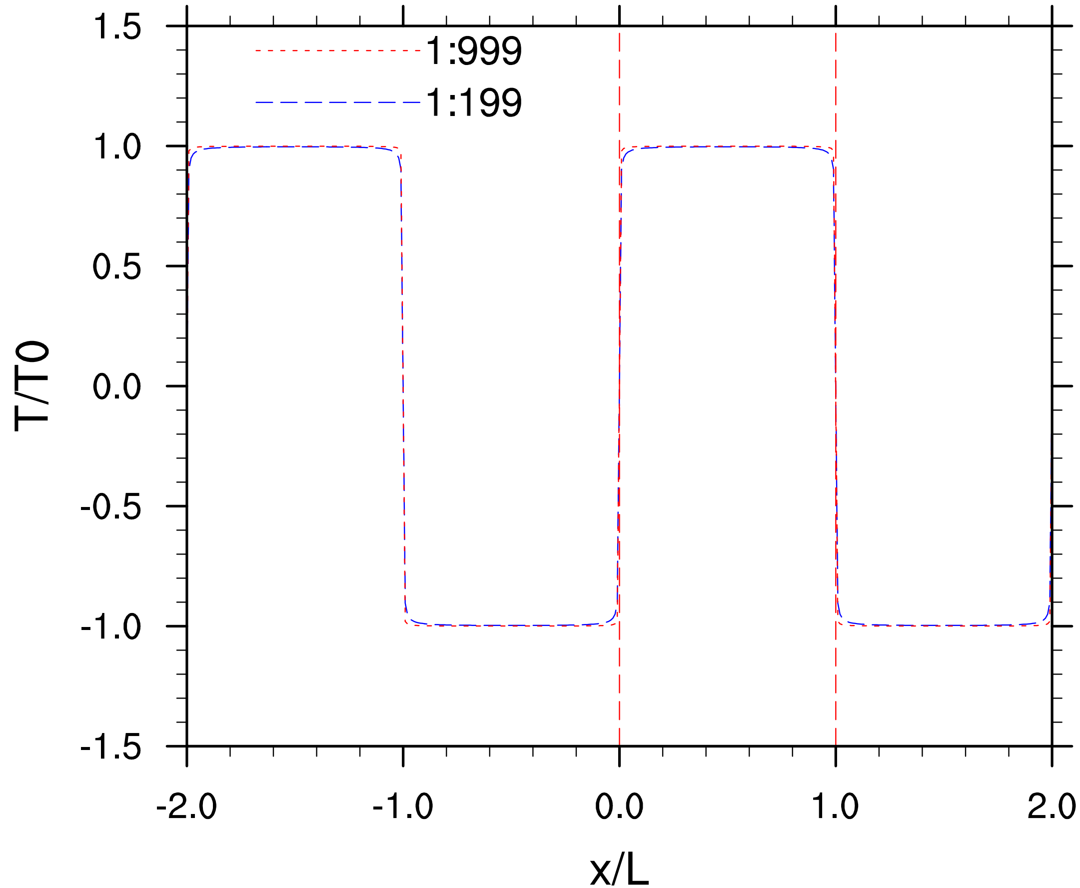
$$T(0,y) = T(L,y) = T(x,L) = 0$$

Analytical solution by the method of separation of variables:

$$T(x,y) = \sum_{n=1,3,5,\ldots} \frac{4T_0}{n\pi \sinh(n\pi)} \sin\left(\frac{n\pi x}{L}\right) \sinh\left(\frac{n\pi(L-y)}{L}\right)$$

$$1 = \sum_{k=1,3,5,\ldots..} \frac{4}{k\pi} \sin\left(k\pi \frac{x}{L}\right)$$

Indeed, in the extended domain, a periodic shape composed of sharp steps is approximated by sine function expansion.



1D Transient heat transfer (Parabolic PDE)

$$\frac{\partial T(x,t)}{\partial t} = \alpha \frac{\partial^2 T(x,t)}{\partial x^2}, \qquad 0 \le x \le L, \qquad t \ge 0$$

Initial condition: $T(x, t=0) = T_0$

Boundary conditions: $T(x=0, t) = T_F, \qquad T(x=L, t) = T_F$

The analytical solution (by the method of separation of variables) is:

$$T(x,t) = T_F + (T_0 - T_F)\sum_{k=1}^{\infty} \frac{4}{\pi(2k-1)} \exp\left(-(2k-1)^2\pi^2\frac{\alpha t}{L^2}\right) \sin\left((2k-1)\pi\frac{x}{L}\right)$$

The normalized form

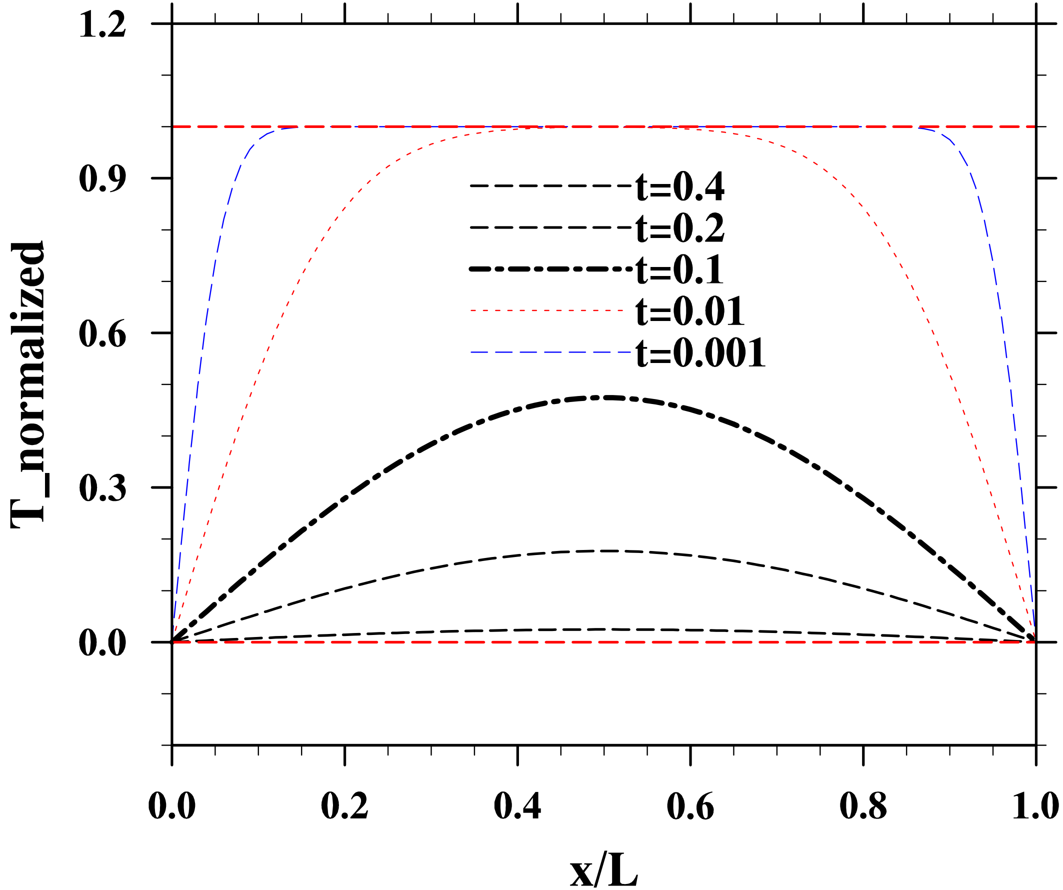
$$\frac{T(x,t) - T_F}{(T_0 - T_F)} = F\left(\frac{x}{L}, \frac{\alpha t}{L^2}\right) = \sum_{k=1}^{\infty} \frac{4}{\pi(2k-1)} \exp\left(-(2k-1)^2\pi^2\frac{\alpha t}{L^2}\right) \sin\left((2k-1)\pi\frac{x}{L}\right)$$

A code to compute the analytical solution: temperature.f90

The normalized form

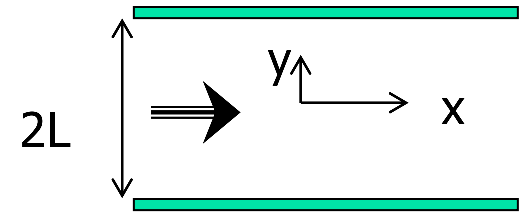
$$\frac{T(x,t)-T_F}{(T_0-T_F)} = F\left(\frac{x}{L},\frac{\alpha t}{L^2}\right) = \sum_{k=1}^{\infty}\frac{4}{\pi(2k-1)}\exp\left(-(2k-1)^2\pi^2\frac{\alpha t}{L^2}\right)\sin\left((2k-1)\pi\frac{x}{L}\right)$$

```
Program HeatDiffusion
 real pi,L,alpha,T0,TF
 real t1,t2,t3,t4,t5,x
 real Tn1,Tn2,Tn3,Tn4,Tn5
 integer k,ix
        pi = 4.*atan(1.0) ! pi=3.14159....
        T0 = 100.0
        TF = 20.0
        alpha = 1.0
        L = 1.0
        t1 = 0.001*alpha/L**2
        t2 = 0.01*alpha/L**2
        t3 = 0.1*alpha/L**2
        t4 = 0.2*alpha/L**2
        t5 = 0.4*alpha/L**2

        do ix = 0,100
        x = 0.01*ix/L
        Tn1 = 0.0
        Tn2 = 0.0
        Tn3 = 0.0
        Tn4 = 0.0
        Tn5 = 0.0
        do k=1,500
        c3 = (2.*k - 1.0)*pi
        c2 = c3*c3
        c1 = 4./c3
        Tn1 = Tn1 + c1*exp(-c2*t1)*sin(c3*x)
        Tn2 = Tn2 + c1*exp(-c2*t2)*sin(c3*x)
        Tn3 = Tn3 + c1*exp(-c2*t3)*sin(c3*x)
        Tn4 = Tn4 + c1*exp(-c2*t4)*sin(c3*x)
        Tn5 = Tn5 + c1*exp(-c2*t5)*sin(c3*x)
        end do

        write(10,100) x, Tn1, Tn2, Tn3, Tn4, Tn5
        end do
100     format(2x,6F12.6)

End Program HeatDiffusion
```



The plot was done by NCL: demo4.ncl

The transient laminar flow in a channel (Parabolic PDE)



$$\vec{u} = (u, v, w)$$

where $u = u(y,t), \qquad v = 0, \qquad w = 0$

Simplified momentum equation

$$\frac{\partial u}{\partial t} = g_x + \frac{\mu}{\rho} \cdot \frac{\partial^2 u}{\partial y^2}$$

$$I.C. \quad u(y, t = 0) = 0$$
$$B.C.s \quad u(y = \pm L, t) = 0$$

Long-time solution implies

$$\rho g_x = \frac{8\mu u_0}{(2L)^2} = \frac{2\mu u_0}{L^2}$$

$u_0 = maximum\ velocity\ at\ y = 0$

1D linear horizontal acoustic wave equation

No viscous friction, isentropic flow [first assumption]

$$p = \rho RT \rightarrow T = \frac{p}{\rho R}, dT = \frac{dp}{\rho R} - \frac{p}{\rho^2 R} d\rho$$

$$ds = \frac{C_v dT}{T} - \frac{R d\rho}{\rho} = \frac{C_v dp}{p} - \frac{C_p d\rho}{\rho} = 0 \quad \rightarrow \quad \frac{p}{p_0} = \left(\frac{\rho}{\rho_0}\right)^\gamma, \qquad \gamma \equiv \frac{C_p}{C_v}$$

Small velocity, small density and pressure fluctuations [second assumption]

$$\frac{p - p_0}{p_0} \ll 1, \qquad \frac{\rho - \rho_0}{\rho_0} \ll 1, \qquad \frac{dp}{p_0} = \gamma \frac{d\rho}{\rho_0}$$

Continuity and momentum equations

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} = 0 \quad \text{linearize} \quad \frac{\partial \rho}{\partial t} + \rho_0 \frac{\partial u}{\partial x} = 0$$

$$\rho \frac{\partial u}{\partial t} + \rho u \frac{\partial u}{\partial x} + \frac{\partial p}{\partial x} = 0 \qquad \text{linearize} \qquad \rho_0 \frac{\partial u}{\partial t} + \frac{\partial p}{\partial x} = 0$$

Eliminate u from the above two

$$\frac{\partial^2 \rho}{\partial t^2} = -\rho_0 \frac{\partial^2 u}{\partial x \partial t} = \frac{\partial^2 p}{\partial x^2} = \frac{p_0 \gamma}{\rho_0} \frac{\partial^2 \rho}{\partial x^2} = c_s^2 \frac{\partial^2 \rho}{\partial x^2}$$

$$\rightarrow \quad \boxed{\frac{\partial^2 \rho}{\partial t^2} - c_s^2 \frac{\partial^2 \rho}{\partial x^2} = 0}$$

Hyperbolic PDF

The speed of sound

$$c_s = \sqrt{\gamma R T_0}$$

The general solution

$$\rho = F_{left}(x + c_s t) + F_{right}(x - c_s t)$$

1D acoustic wave in unbounded domain

$$\frac{\partial^2 \rho}{\partial t^2} - c_s^2 \frac{\partial^2 \rho}{\partial x^2} = 0, \qquad -\infty < x < +\infty, \qquad t \geq 0$$

Initial conditions: $\rho(x,0) - \rho_0 = f(x); \quad \frac{\partial}{\partial t}[\rho(x,0) - \rho_0] = g(x)$

The solution is:

$$\rho(x,t) - \rho_0 = \frac{1}{2}[f(x + c_s t) + f(x - c_s t)] + \frac{1}{2c_s}\int_{x-c_s t}^{x+c_s t} g(\tau)\ d\tau$$

At any given time, the solution at x_p is determined by the initial data within

$$x_p - c_s t \leq x \leq x_p + c_s t$$

Types of equations in CFD (Computational Fluid Dynamics)

Laplace Equation

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$$

Poisson Equation

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = \mathrm{S}(x, y)$$

Laplace equations and Poisson equations are elliptic equations and generally associated with the steady-state problems.

The velocity potential in steady, inviscid, incompressible, and irrotational flows satisfied the Laplace equation.

The pressure field is, in general, governed by the Poisson equation.

The temperature distribution for steady-state, constant-property, two-dimensional condition satisfies the Laplace equation if no volumetric heat source is present.

Fluid flow problems

- generally have nonlinear terms due to the inertia or acceleration
- component in the momentum equation. These terms are called advection terms. The energy equation has nearly similar terms, usually called the convection terms, which involve the motion of the flow field. For unsteady two-dimensional problems, the appropriate equation can be represented as

$$\frac{\partial \phi}{\partial t} + u\frac{\partial \phi}{\partial x} + v\frac{\partial \phi}{\partial y} = B\left(\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2}\right) + S$$

- ϕ denotes velocity, temperature or some other transported property,
- u and v are velocity components,
- B is the diffusivity for momentum or heat,
- S is a source term.

The pressure gradients in the momentum or the volumetric heating in the energy equation can be appropriately substituted in S.

The above PDE is parabolic in time and elliptic in space.

For very high-speed flows, the terms on the left side dominate, the second-order terms on the right hand side become trivial, and the equation become hyperbolic in time and space.

Summary

- With respect to two independent variables, PDE can be classified into
  - Elliptic: steady-state heat diffusion in 2D
  - Parabolic: unsteady heat diffusion, 1D unsteady channel flow
  - Hyperbolic: acoustic wave equation, inviscid compressible flow

  Their solutions have different characteristics.

  Examples of each type are provided

- Linear PDE can often be solved analytically by the method of separation of variables

- The Navier-Stokes equation contains more than two independent variables as well as more than one unknown fields, the nature of PDE depends on which two independent variables are selected, or whether the flow is inviscid or not. The nature may also change from one spatial region to another. For some special cases, the nature can be shown explicitly.

- The Navier-Stokes equation is a treasure box which has all types of PDE.